

Sri Lanka Institute of Information Technology



IT3091 - Machine Learning

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Group Assignment - Initial Report

**Global Logistics: Proactive Supply Chain Delay Prevention
and
Bottleneck Diagnostics**

Group Number: 2026-AI-58

Name	Student ID Number
Lithman P.V. L	IT24101075
Gangoda G.M.M.D.B	IT24100743
Irham M.I.M	IT24103811

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01.Introduction

1.1 Project Overview

Global Logistics: Proactive Supply Chain Delay Prevention and Bottleneck Diagnostics is a machine learning and data mining project focused on supporting data-driven operational decisions within global supply chain management. The project utilizes the DataCo Smart Supply Chain dataset, which captures detailed transaction-level e-commerce order and fulfillment data across international logistics networks.

The project aims to develop an early-warning Machine Learning classification architecture capable of accurately predicting order delivery delay risks at the exact moment of checkout, combined with an Unsupervised Data Mining model to diagnose macro structural bottlenecks. The raw dataset contains 180,519 transaction records and 53 columns, including logistical, geographical, financial, and order-related attributes alongside the primary target variable, `Late_delivery_risk`.

Before model development, the dataset will be systematically examined through data understanding and exploratory data analysis (EDA) to identify key operational patterns, high-risk delivery corridors, variable relationships, and post-fulfillment data leakage features that must be removed. These findings will guide the subsequent data preprocessing, feature engineering, and model validation stages.

The expected outcome is a reliable, evidence-based proactive decision-support system that enables logistics operations managers to transition from reactive delay management to automated early risk mitigation at order placement.

1.2 Business Context

The global logistics and retail e-commerce industry requires predictable transit led times to maintain Service Level Agreement (SLA) compliance and secure customer trust. A global enterprise needs to anticipate whether an incoming customer order will experience a shipping delay based on checkout attributes such as shipping mode, destination market, product category, order volume, and scheduled transit days.

In practice, standard logistics operations involve managing thousands of daily cross-border transactions across diverse carrier networks, making it impossible to consistently detect delay risks using manual oversight alone. Analysis of baseline operations reveals that over 54% of total shipments suffer from late delivery risk. Currently, management relies on reactive workflows—discovering delays only *after* the scheduled delivery date has lapsed.

This reactive failure mode leads to severe business consequences:

- **Financial Penalties:** Substantial SLA breach fines and excessive emergency freight costs incurred to expedite delayed packages.
- **Customer Churn:** Reduced customer lifetime value (CLV) and eroded brand reputation resulting from unfulfilled delivery promises.

To solve this, the project applies a dual decision-lens machine learning approach:

1. **Primary Lens (Real-Time Delay Prevention):** Uses Supervised Binary Classification to score incoming orders at checkout, outputting a real-time risk probability to trigger immediate warehouse picking prioritization or carrier upgrades.
2. **Secondary Lens (Systemic Bottleneck Diagnostics):** Uses Unsupervised K-Means Clustering to group shipping corridors and carrier modes into structural risk profiles, assisting operations leaders in long-term SLA adjustments and contract renegotiation.

1.3 Dataset Overview

The dataset used for this project is the **DataCo Smart Supply Chain** dataset. It contains 180,519 order line-item instances and 53 columns, with `Late_delivery_risk` (0 = On Time, 1 = Late) serving as the primary binary target variable for classification.

The dataset comprises a mix of numerical and categorical variables describing various operational dimensions of international retail logistics:

- **Logistical & Transit Features:** Scheduled shipping days, actual shipping days, shipping mode, delivery status, and late delivery risk flag.
- **Geographical Attributes:** Destination Market (e.g., LATAM, Europe, USCA), Order Region, Order Country, Order City, and Customer State.
- **Commercial & Financial Metrics:** Product price, item discount amount, order item total, profit ratio, sales volume, category name, and department.
- **Temporal & Customer Data:** Order transaction timestamps, order hour, order day of the week, customer segment, and fulfillment status.

02. Business Problem & Analytical Lens

2.1 Business Problem

The global logistics organization faces severe operational challenges due to unpredictable transit delays across its international retail fulfillment networks. Analysis of the DataCo transaction records reveals that over 54% of total orders suffer from late delivery risk under default operational workflows.

Determining whether an order will be delivered late at the exact moment of checkout is highly challenging because transit lead times are governed by a complex mix of order attributes, including shipping mode selections, cross-border customs constraints, destination market infrastructure, picking/packing lead times, and product category characteristics.

Under current operational workflows, logistics managers rely on **reactive management**—discovering delivery delays only *after* scheduled Service Level Agreement (SLA) deadlines have already lapsed. At this late stage, corrective actions are expensive and largely ineffective, leading to severe business consequences:

- **Financial Losses:** Heavy SLA breach penalties and excessive emergency freight costs incurred to expedite delayed shipments.
- **Customer Churn:** Eroded customer trust, decreased Customer Lifetime Value (CLV), and negative brand reputation resulting from unfulfilled delivery promises.

Therefore, the company requires a data-driven proactive decision-support architecture to predict order delay risks instantly at checkout and diagnose structural supply chain bottlenecks across its global shipping network.

2.2 Stakeholder and Decision Need

No	Stakeholder	Decision need
<i>Primary Stakeholders</i>		
1	Logistics Operations Managers & Warehouse Dispatchers	To receive real-time order-level delay risk predictions at checkout, using the score to prioritize warehouse picking queues, re-route shipments, or automatically upgrade carrier service classes prior to package dispatch
<i>Secondary Stakeholders</i>		
1	Chief Logistics Officer (CLO) & Supply Chain Executives	To evaluate structural bottleneck diagnostics across transit corridors and carrier modes, using the insights to renegotiate carrier contracts, adjust regional SLAs, and optimize inventory re-allocation.
2	Customer Service & Support Team	To access early-warning delivery risk flags, enabling proactive customer communication and expectation

		management before SLA breaches occur.
3	Carrier Network Partners (3PL Logistics)	To identify specific high-risk transit corridors and destination markets that consistently exceed scheduled shipping windows, supporting joint capacity planning and infrastructure improvements

2.3 Primary Analytical Lens.

The primary analytical lens of this project is **Operational Delay Prevention** (Real-Time Predictive Risk Lens).

Operational Delay Prevention was selected as the primary analytical lens because the logistics organization must intervene *before* physical package dispatch to prevent SLA breaches. Once an order enters transit under a standard carrier class, logistics managers have minimal control over transit times. The moment of order placement at checkout represents the critical operational window to apply risk mitigation actions.

Machine learning can identify non-linear relationships and interactions between checkout attributes (e.g., Shipping Mode, Market, Order Region, Category Name, Order Item Total, and Days for shipment (scheduled)) and historical delivery outcomes. Supervised binary classification models can process these attributes to output a real-time risk probability score (0.0 to 1.0) for the target variable `Late_delivery_risk` (1 = Late, 0 = On Time). Therefore, supervised binary classification provides the most appropriate technical approach for driving real-time operational delay prevention.

2.4 Secondary Analytical Lens.

The secondary analytical lens of this project is **Systemic Bottleneck Diagnostics** (Macro Structural Analysis).

Systemic Bottleneck Diagnostics was selected as the secondary analytical lens to complement order-level predictions with macro-level root-cause diagnostics. While the primary lens focuses on predicting individual order outcomes, operations executives also require a structural understanding of *why* and *where* delays cluster across the global supply chain network.

Using unsupervised **K-Means Clustering**, order attributes and shipping corridors (combinations of shipping mode, market, and destination region) are aggregated and segmented based on their operational delay profiles (e.g., transit lead time variance, late delivery proportions, and order volumes). This groups supply chain lanes into distinct risk clusters (e.g., *Severe Bottleneck Corridors* vs. *High-Reliability Routes*), enabling executive stakeholders to address root-cause failures through long-term SLA recalibration and carrier contract renegotiation

03. Proposed Machine Learning Task & Expected Output

3.1 Proposed Machine Learning Task

The primary machine learning task is **Supervised Binary Classification** because the project aims to predict a discrete binary operational outcome, `Late_delivery_risk`, using checkout-time transaction features in the structured dataset. The dataset contains labelled historical order records where the true delivery outcome (Late vs. On-Time) is known.

Additionally, an **Unsupervised K-Means Clustering** task is integrated as a secondary diagnostic model to cluster macro shipping corridors into distinct operational risk profiles based on structural delay metrics.

3.2 Target Variable and Unit of Analysis

- **Target Variable:** The primary target variable is `Late_delivery_risk`, represented as a binary outcome:
 - 1 = Late Delivery Risk Detected (Actual delivery days exceeded scheduled SLA days)
 - 0 = On-Time Delivery Expected (Delivery completed on or before scheduled SLA days)
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- **Unit of Analysis:** The unit of analysis is an **individual order line item** (Order Item Id). Each row in the dataset represents a single item transaction within an e-commerce order along with its associated logistical, geographical, temporal, and commercial characteristics.

3.3 Expected Main Output

The main output of the primary machine learning model is a **real-time risk probability score** (0.0 to 1.0) generated for each order line item at checkout. This continuous probability score measures the likelihood of an SLA breach, allowing the Warehouse Management System (WMS) to flag high-risk orders and trigger automated operational interventions (e.g., priority picking or shipping mode upgrades) prior to physical package dispatch.

3.4 Expected Supporting Insights

the project is also expected to provide two key layers of supporting diagnostic insights:

1. **Model Interpretability & Feature Importance (SHAP Analysis):** Identifying the specific checkout-time order characteristics (e.g., Shipping Mode, Days for shipment (scheduled), Market, Order Region, Category Name) that contribute most significantly to late delivery predictions.
2. **Structural Bottleneck Clustering:** Segmenting international supply chain lanes into macro-operational risk clusters (e.g., *Severe Bottleneck Corridors*, *Moderate Delay Corridors*, and *High-Reliability Routes*) to support executive decision-making regarding carrier contract renegotiations and regional SLA recalibration

04. Proposed Workflow

DataCo Machine Learning Project Workflow



05. Group Member Responsibilities.

Member	IT Number	Responsibility	Main Tasks
Gangoda G.M.M.D.B	IT24100743	Problem Definition, EDA & Leakage Elimination	Business problem & dual decision lens formulation • Dataset understanding & data quality analysis • Exploratory Data Analysis (EDA) on operational delay trends • Identification and complete removal of post-fulfillment target leakage features • EDA & Data Leakage Insight Log
Irham M.I.M	IT24103811	Data Preprocessing, Feature Engineering & Unsupervised Clustering	• Stratified random train/test split (80/20) • Feature engineering (SLA ratios, temporal order extractions) • Categorical encoding & Column Transformer pipeline construction • Secondary Lens: Shipping corridor aggregation • Unsupervised K-Means clustering, Elbow Method & Silhouette analysis
Lithman P.V. L	IT24101075	Classification Modeling, SHAP Diagnostics & Operational Recommendations	• Primary Lens: Supervised classification model development (Logistic Regression, Decision Tree, Random Forest, LightGBM/XGBoost)

			• 5-Fold Stratified Cross-Validation & evaluation metrics (Recall, ROC-AUC, Precision) • Hyperparameter tuning & decision threshold optimization • Model explainability (SHAP Analysis) • Operational Action Matrix & stakeholder recommendations
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Common Group Responsibilities: All members will collaboratively review and integrate their respective work, verify the results, maintain reproducibility, prepare the final report and demonstration, ensure AI-use transparency, and complete the final submission requirements. Each member will also prepare their individual Personal Learning Journey.

06. Expected Project Contribution & Business Value

6.1 Expected Contribution

This project is expected to develop an end-to-end Machine Learning and Data Mining architecture for proactive supply chain delay prevention and systemic bottleneck diagnostics using the DataCo Smart Supply Chain dataset.

The primary contribution is a **Supervised Binary Classification pipeline** capable of accurately predicting `Late_delivery_risk` at the exact moment of order placement. The project will benchmark multiple classification algorithms—including Logistic Regression, Decision Tree, Random Forest, and LightGBM/XGBoost—evaluated through 5-Fold Stratified Cross-Validation with a primary focus on **Recall** and **ROC-AUC** metrics.

Additionally, the project contributes an **Unsupervised K-Means Clustering model** to aggregate macro shipping corridors and segment supply chain routes into distinct operational risk profiles. Through SHAP (SHapley Additive exPlanations) analysis, the project will extract key operational feature importances, providing interpretability regarding the specific order and transit characteristics that drive delivery delays

6.2 Expected Stakeholder Value

he proposed solution is expected to transform global logistics operations by shifting organizational workflows from **reactive firefighting** to **proactive risk mitigation**:

- **Logistics Operations Managers & Warehouse Dispatchers:** Benefit from real-time delay probability scores (0.0 to 1.0) generated at checkout. High-risk orders can automatically be assigned priority picking tags in the warehouse queue or routed to faster shipping carriers prior to physical package dispatch, preventing SLA breaches.
- **Chief Logistics Officer & Supply Chain Executives:** Gain strategic insights from the K-Means corridor clustering and SHAP diagnostics. Executive stakeholders can identify structural failure patterns across specific shipping modes and regional markets, enabling evidence-based carrier contract renegotiations and long-term SLA recalibrations.
- **Customer Support & Service Teams:** Receive early-warning risk flags to proactively manage customer expectations before delivery deadlines lapse, protecting Customer Lifetime Value (CLV) and brand trust.

6.3 Expected Limitations

The project is expected to have several operational and data-driven limitations:

- **Dataset & Temporal Scope:** The model is trained on historical transactions within the DataCo dataset, which may not fully capture sudden macroeconomic shifts, global supply chain disruptions, or seasonal demand spikes in specific regional markets.

- **Absence of Real-Time External Features:** The dataset lacks external real-time operational variables—such as live weather disruptions, local traffic congestion, port strikes, or customs clearance backlogs—which heavily influence actual transit times.
- **Checkout Feature Constraint:** Due to strict target leakage elimination (removing post-fulfillment variables like actual shipping days and shipment status), the classification model relies exclusively on attributes available at checkout, which may constrain maximum predictive accuracy.

Consequently, the model's outputs should be treated as **probabilistic decision-support scores** rather than deterministic guarantees, designed to optimize operational resource allocation alongside human oversight.

07. Conclusion

This project proposal presents a dual decision-lens Machine Learning and Data Mining solution to resolve unpredictable delivery delays within global retail supply chains using the DataCo Smart Supply Chain dataset. By combining **Operational Delay Prevention** (Primary Supervised Classification Lens) with **Systemic Bottleneck Diagnostics** (Secondary Unsupervised Clustering Lens), the architecture addresses both real-time order-level risks and long-term structural supply chain friction.

The proposed workflow covers the end-to-end project lifecycle: business framing, target leakage elimination, exploratory data analysis, feature engineering, stratified 80/20 data splitting, 5-fold cross-validation model benchmarking, K-Means route clustering, SHAP model interpretability, and actionable stakeholder recommendations. By prioritizing Recall to minimize costly missed delays, the final model provides global logistics managers with a data-driven framework to reduce SLA penalties, optimize warehouse operations, and improve customer satisfaction